EIGHT EQUIVALENT CONDITIONS FOR THE VANISHING OF EXT GROUPS OUT OF A LOCALIZATION AT A SINGLE ELEMENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The last enumerated item of the statement, 'add more here.', is the author's own placeholder in the pinned source; it is delivered verbatim rather than removed, and it contributes no mathematical condition. The footnote in the counted proof span, which explains the unbounded-below case and ends 'Details omitted.', is likewise delivered verbatim; that footnote records the author applying the argument to a replacement double complex rather than invoking the referenced equation, and it is part of the counted proof span. The author's typo 'we chooses' is reproduced unchanged.

Editorial note. The references inside the counted proof span are recorded as non-load-bearing navigation pointers: algebra-lemma-adjoint-hom-restrict, derived-lemma-K-injective, derived-section-derived-limit and homology-section-double-complex resolve to the project's cross-chapter tag list, while lemma-hom-K-injective and equation-first-ss-ext are same-chapter labels.

Allowed standard prerequisites (editorial).

- K-injective complexes and the fact that an acyclic K-injective complex is homotopy equivalent to zero. used at proof lines 25870-25884. Is the device that turns the vanishing of Ext groups into the vanishing of maps in the derived category.
- The adjunction identifying Hom over A from A_f with Hom over A_f after applying Hom_A(A_f, -). used at proof lines 25875-25879. Lets a map out of an A_f-complex be rewritten as a map into Hom_A(A_f, I^bullet).
- The explicit two-term free resolution of A_f and the construction of derived limits. written out at proof lines 25887-25895 and invoked at proof lines 25902-25905. Supplies the displayed pro-product map in condition (7) and the identification of T(K, f) with R Hom_A(A_f, K).
- Ext spectral sequence of a two-term projective resolution and its degeneration. used at proof lines 25909-25925. Produces the short exact sequences

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

that relate Ext^p_A(A_f, K) to Hom and Ext^1 of the cohomology modules.

Source reference. Original lemma 08YP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11347–11347. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 070I: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8957–8957. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08TB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11159–11159. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original equation 17370: sources/stacks-more-algebra/pinned-original-source.tex, source lines 17370–17370. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 012X: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1415–1415. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 13956: sources/stacks-more-algebra/pinned-original-source.tex, source lines 13956–13956. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let A be a ring. Let $f \in A$. Let $K \in D(A)$. The following are equivalent*

1. $\mathrm{Ext}^n_A(A_f, K) = 0$ *for all* n,
2. $\mathrm{Hom}_{D(A)}(E, K) = 0$ *for all* E *in* $D(A_f)$,
3. $T(K, f) = 0$,
4. *for every* $p \in \mathbf{Z}$ *we have* $T(H^p(K), f) = 0$,
5. *for every* $p \in \mathbf{Z}$ *we have* $\mathrm{Hom}_A(A_f, H^p(K)) = 0$ *and* $\mathrm{Ext}^1_A(A_f, H^p(K)) = 0$,
6. $R\mathrm{Hom}_A(A_f, K) = 0$,
7. *the map* $\prod_{n\geq 0} K \to \prod_{n\geq 0} K$, $(x_0, x_1, \ldots) \mapsto (x_0 - fx_1, x_1 - fx_2, \ldots)$ *is an isomorphism in* $D(A)$, *and*
8. *add more here.*

Proof. It is clear that (2) implies (1) and that (1) is equivalent to (6). Assume (1). Let $I^\bullet$ be a K-injective complex of A-modules representing K. Condition (1) signifies that $\mathrm{Hom}_A(A_f, I^\bullet)$ is acyclic. Let $M^\bullet$ be a complex of A_f-modules representing E. Then

$$\mathrm{Hom}_{D(A)}(E, K) = \mathrm{Hom}_{K(A)}(M^\bullet, I^\bullet) = \mathrm{Hom}_{K(A_f)}(M^\bullet, \mathrm{Hom}_A(A_f, I^\bullet))$$

by Algebra, Lemma 08YP. As $\mathrm{Hom}_A(A_f, I^\bullet)$ is a K-injective complex of A_f-modules by Lemma 13956 the fact that it is acyclic implies that it is homotopy equivalent to zero (Derived Categories, Lemma 070I). Thus we get (2).

A free resolution of the A-module A_f is given by

$$0 \to \bigoplus\nolimits_{n\in\mathbf{N}} A \to \bigoplus\nolimits_{n\in\mathbf{N}} A \to A_f \to 0$$

where the first map sends the $(a_0, a_1, a_2, \ldots)$ to $(a_0, a_1 - fa_0, a_2 - fa_1, \ldots)$ and the second map sends $(a_0, a_1, a_2, \ldots)$ to $a_0 + a_1/f + a_2/f^2 + \ldots$. Applying $\mathrm{Hom}_A(-, I^\bullet)$ we get

$$0 \to \mathrm{Hom}_A(A_f, I^\bullet) \to \prod I^\bullet \to \prod I^\bullet \to 0$$

Since $\prod I^\bullet$ represents $\prod_{n \geq 0} K$ this proves the equivalence of (1) and (7). On the other hand, by construction of derived limits in Derived Categories, Section 08TB the displayed exact sequence shows the object $T(K, f)$ is a representative of $R\mathrm{Hom}_A(A_f, K)$ in $D(A)$. Thus the equivalence of (1) and (3).

There is a spectral sequence

$$E_2^{p,q} = \mathrm{Ext}_A^p(A_f, H^q(K)) \Rightarrow \mathrm{Ext}_A^{p+q}(A_f, K)$$

See Equation (17370)[1]. This spectral sequence degenerates at E_2 because A_f has a length 1 resolution by projective A-modules (see above) hence the E_2-page has only 2 nonzero columns. Thus we obtain short exact sequences

$$0 \to \mathrm{Ext}_A^1(A_f, H^{p-1}(K)) \to \mathrm{Ext}_A^p(A_f, K) \to \mathrm{Hom}_A(A_f, H^p(K)) \to 0$$

This proves (4) and (5) are equivalent to (1). □

[1]If K is not bounded below, we can't use the given reference. In that case, we chooses a complex $K^\bullet$ representing K and a free two-term resolution $0 \to F^{-1} \to F^0 \to A_f \to 0$ (see above). Then we get a double complex with terms $\mathrm{Hom}_A(F^{-q}, K^p)$ whose totalization computes $R\mathrm{Hom}(A_f, K)$. The second spectral sequence associated to this double complex (Homology, Section 012X) gives us what we want. Details omitted.